S&P Global

Crisil
a company of S&P Global

CODE TO CONNECT
S&P Global & Crisil | Hackathon 2026

AI/NLP Risk Engine

with a Sentiment Index Rebalancer and an Event-Driven Stress Test

Sarthak Satish Borekar

B.Tech CSE, VIT Bhopal University | Reg. 23BCE10568

S&P Global & Crisil Campus Hackathon 2026 | Individual submission

Problem and approach

Too much text, too little time: turn it into signals a risk team can act on

The problem

- News and social posts arrive faster than anyone can read.
- Free text cannot feed a portfolio system directly.
- Risk and portfolio teams need a score, an event type and a severity, per document.

Core engine output per document

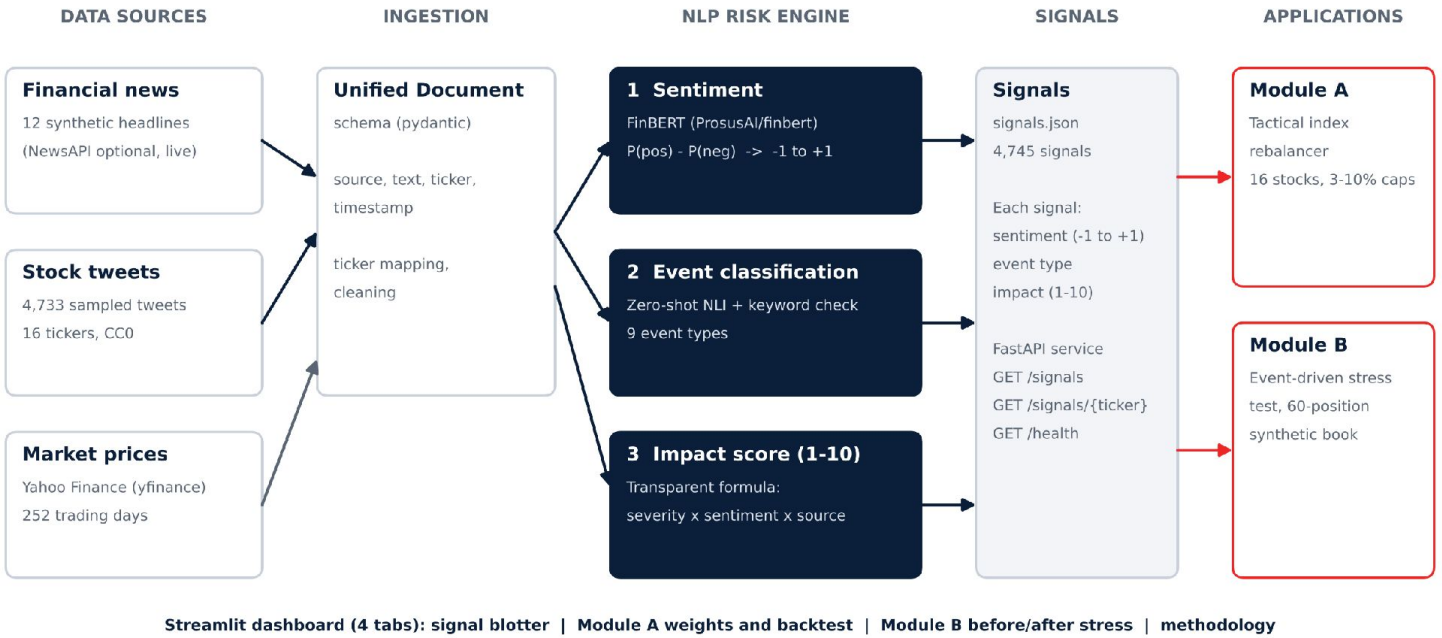
Sentiment	Event	Impact
-1 to +1	9 types	1 to 10

Our approach

1. **Ingest**
   News headlines and stock tweets into one document schema
2. **Score**
   FinBERT sentiment, zero-shot event type, formula-based impact
3. **Apply**
   Module A: tactical index rebalancer
   Module B: stress test of a synthetic bank book
4. **Show**
   API (FastAPI), JSON file and a Streamlit dashboard

System design

Two text sources in, one signal store, two applications on top



Implementation highlights

Three scores per document, each simple enough to explain

Sentiment

FinBERT (ProsusAI/finbert)

Score = P(positive) - P(negative), so the result sits between -1 and +1. A model trained on financial text, not generic reviews.

Event type

Zero-shot NLI + keyword check

Nine labels from Credit Event to Other. A systemic label needs a keyword hit, and weak guesses with no keyword support become Other.

Impact 1-10

Transparent formula

1 + 9 x severity x (0.4 + 0.6|sentiment|) x source weight x (0.6 + 0.4 x event confidence). No labeled impact data exists, so weights are stated assumptions.

84.4% event accuracy on 45 test headlines (keyword 88.9%, zero-shot 77.8%). All 7 errors were non-systemic: no false Credit, Geopolitical or Macro calls. Headlines were written by me with AI help, so treat as indicative.

Key results

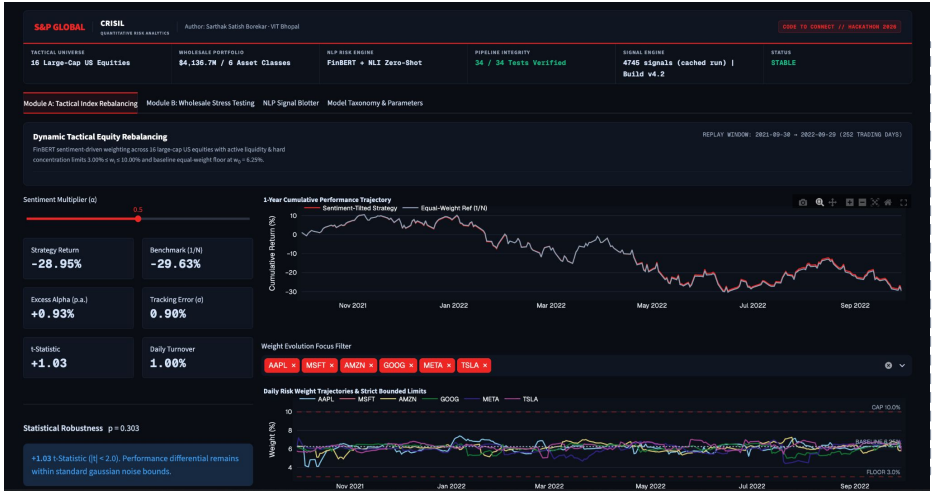
Real outputs from 4,745 signals (12 headlines + 4,733 tweets, 16 stocks)

Module A: sentiment-tilted index, 252-day replay

-28.95%
Sentiment index return

-29.63%
Equal-weight benchmark

t = 1.03
+0.93%/yr, not significant





Key results

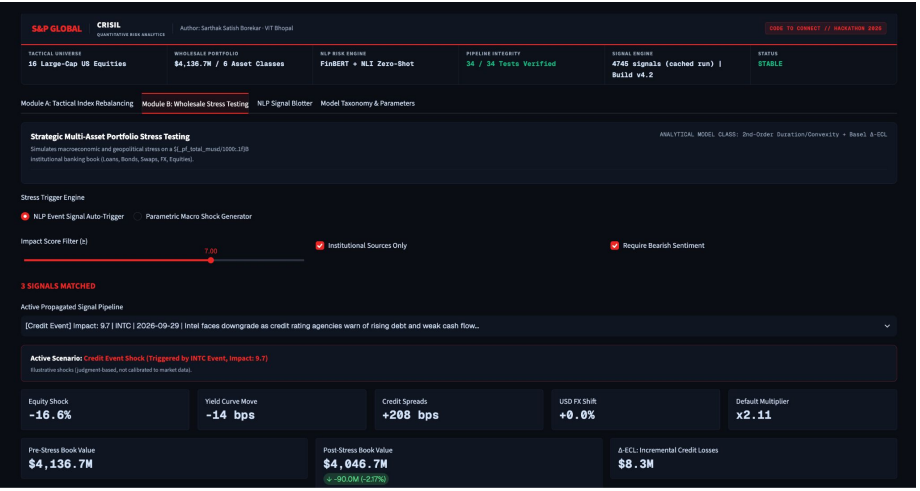
Real outputs from 4,745 signals (12 headlines + 4,733 tweets, 16 stocks)

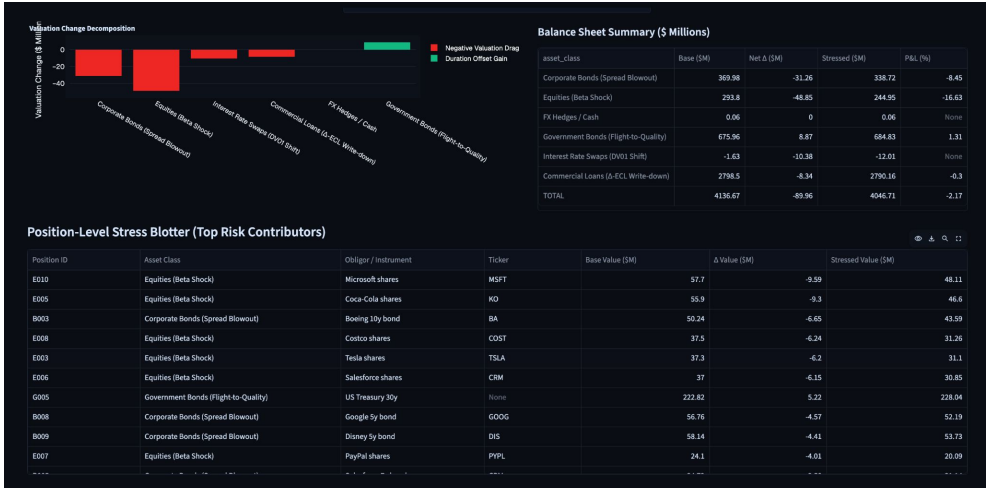
Module B: stress test on a 60-position synthetic book

-$90.0M
Credit Event, impact 9.7

-2.17%
$4,136.7M to $4,046.7M

+$8.3M
Extra expected loan loss





Domain impact

From unstructured text to decisions a bank or asset manager already makes

1 **Triage**

Filter the feed down to the few high-impact items instead of reading every headline. Only 3 of 12 headlines trigger a stress test.

2 **Portfolio overlay**

Sentiment tilts index weights inside 3%-10% limits, with weights that always sum to 100% and no look-ahead in the replay.

3 **Stress testing**

A systemic event (Credit, Geopolitical, Macro) with high impact and negative tone shocks equities, rates, spreads, USD and PDs across loans, bonds, swaps and FX.

Limitations and next steps

What this prototype does and does not claim

Limitations

- No proven return edge: t = 1.03, not significant
- Impact weights and stress shocks are assumptions, not calibrated
- Tweet coverage is uneven (TSLA far above JNJ, GS)
- Batch run on cached signals, not real-time streaming

Next steps

- Fine-tune sentiment and event models on labeled financial text
- Learn impact from historical market reactions
- Calibrate stress shocks to historical episodes
- Add live streaming ingestion and counterparty risk